

# Chapter 6: Jointly Distributed Random Variables (Part 2)

## Sections Covered:

### 6.3 Conditional Distributions

### 6.4 Conditional Expectation

### 6.5 Sums of Random Variables

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## Big Picture

In Part 1, we studied:

- Joint distributions
- Marginal distributions
- Independence

Now we move to an equally important question:

What happens when we already know something about one random variable?

For example:

- What is the distribution of a student's marks given that they studied 10 hours?
- What is the lifetime of a machine given that it has already survived 5 years?
- What is the value of  $Y$  given that  $X = x$ ?

This leads to the concept of **conditional distributions**.

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## 6.3 Conditional Distributions

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### Motivation

Suppose two random variables  $X$  and  $Y$  are jointly distributed.

If we learn that

$$Y = y$$

then our knowledge about  $X$  changes.

We no longer use the ordinary distribution of  $X$ .

Instead, we use the **conditional distribution of  $X$  given  $Y = y$** .

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### Conditional PMF (Discrete Case)

Suppose  $X$  and  $Y$  are discrete.

The conditional probability that  $X = x$  given  $Y = y$  is

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

provided

$$P(Y = y) > 0$$

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Using PMFs,

$$p_{X|Y}(x|y) = \frac{p(x, y)}{p_Y(y)}$$

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### Interpretation

Think of it as:

Restrict attention only to outcomes where  $Y = y$ .

Then renormalize probabilities so that they add up to 1.

## Example

Joint PMF:

| X | Y | Probability |
|---|---|-------------|
| 1 | 1 | 0.2         |
| 1 | 2 | 0.3         |
| 2 | 1 | 0.1         |
| 2 | 2 | 0.4         |

Find

$$P(X = 1|Y = 2)$$

First compute

$$P(Y = 2) = 0.3 + 0.4 = 0.7$$

Then

$$P(X = 1, Y = 2) = 0.3$$

Therefore

$$P(X = 1|Y = 2) = \frac{0.3}{0.7} = \frac{3}{7}$$

## Properties of Conditional PMF

For fixed  $y$ ,

$$p_{X|Y}(x|y) \geq 0$$

and

$$\sum_x p_{X|Y}(x|y) = 1$$

Thus it is a legitimate probability distribution.

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## Conditional Density (Continuous Case)

Suppose  $X$  and  $Y$  have joint density

$$f(x, y)$$

and marginal density

$$f_Y(y)$$

Then

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

whenever

$$f_Y(y) > 0$$

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Similarly,

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)}$$

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## Interpretation

Conditional density tells us:

How the distribution of  $X$  behaves after fixing  $Y = y$ .

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## Independence and Conditional Distributions

Suppose  $X$  and  $Y$  are independent.

Then

$$f(x, y) = f_X(x)f_Y(y)$$

Substituting,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f_X(x)f_Y(y)}{f_Y(y)} \\ &= f_X(x) \end{aligned}$$

Thus

$$f_{X|Y}(x|y) = f_X(x)$$

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## Extremely Important Result

If  $X$  and  $Y$  are independent,

knowing  $Y$  gives absolutely no information about  $X$ .

Therefore

$$P(X = x|Y = y) = P(X = x)$$

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## Conditional Distribution Function

Sometimes we want the entire conditional CDF.

For discrete or continuous variables,

$$F_{X|Y}(x|y) = P(X \leq x|Y = y)$$

This completely describes the conditional distribution.

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## 6.4 Conditional Expectation

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# Motivation

Suppose we know

$$Y = y$$

and want the expected value of  $X$ .

Instead of using

$$E[X]$$

we use

$$E[X|Y = y]$$

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This is called the **conditional expectation**.

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## Discrete Definition

If  $X$  is discrete,

$$E[X|Y = y] = \sum_x x P(X = x|Y = y)$$

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Interpretation:

Take the expectation using the conditional distribution instead of the ordinary distribution.

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## Example

Suppose

$$P(X = 1|Y = 2) = \frac{3}{7}$$

$$P(X = 2|Y = 2) = \frac{4}{7}$$

Then

$$\begin{aligned} E[X|Y = 2] &= 1 \cdot \frac{3}{7} + 2 \cdot \frac{4}{7} \\ &= \frac{11}{7} \end{aligned}$$

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## Continuous Definition

If  $X$  is continuous,

$$E[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

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## Conditional Expectation is a Function

Notice something important.

The quantity

$$E[X|Y = y]$$

depends on  $y$ .

Therefore it is actually a function of  $y$ .

Let

$$g(y) = E[X|Y = y]$$

Then

$$E[X|Y] = g(Y)$$

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## Why This Is Important

Instead of being a number,

$$E[X|Y]$$

is itself a random variable.

This idea becomes central in:

- Regression
  - Machine Learning
  - Bayesian Statistics
  - Stochastic Processes
  - Financial Mathematics
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## Example

Suppose

$$E[X|Y = y] = 2y + 1$$

Then

$$E[X|Y] = 2Y + 1$$

which is itself random because  $Y$  is random.

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## Law of Total Expectation

One of the most important results in probability.

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### Statement

$$E[E(X|Y)] = E(X)$$

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## Meaning

You can compute an expectation in two stages:

### Step 1

Condition on  $Y$

$$E(X|Y)$$

## Step 2

Take expectation again

$$E[E(X|Y)]$$

The answer is exactly

$$E(X)$$

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## Intuition

Conditioning merely reorganizes the information.

It does not change the overall average.

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## Discrete Version

$$E(X) = \sum_y E(X|Y = y)P(Y = y)$$

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## Continuous Version

$$E(X) = \int E(X|Y = y)f_Y(y) dy$$

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## Law of Total Variance (Preview)

A very important identity later in probability:

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

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Interpretation:

Total variation equals

- Average variation remaining after knowing  $Y$
  - Plus variation explained by  $Y$
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## 6.5 Sums of Random Variables

One of the most useful topics in probability.

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### Motivation

Many random quantities are sums.

Examples:

- Total profit
- Total waiting time
- Total rainfall
- Total insurance claims
- Total number of arrivals

Thus we often need the distribution of

$$X + Y$$

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### Distribution of a Sum (Discrete Case)

Suppose

$$Z = X + Y$$

We want

$$P(Z = z)$$

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Since

$$Z = z$$

means

$$X + Y = z$$

we sum over all possible combinations.

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## Fundamental Formula

$$P(Z = z) = \sum_x P(X = x, Y = z - x)$$

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Using joint PMF,

$$p_Z(z) = \sum_x p(x, z - x)$$

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## Example

Let

$$X, Y$$

be independent fair dice.

Find

$$P(X + Y = 7)$$

Possible pairs:

$$(1, 6)$$

$$(2, 5)$$

$$(3, 4)$$

$$(4, 3)$$

$$(5, 2)$$

$$(6, 1)$$

Each has probability

$$\frac{1}{36}$$

Thus

$$P(X + Y = 7) = 6 \left( \frac{1}{36} \right) = \frac{1}{6}$$

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## Convolution Formula (Discrete)

If  $X$  and  $Y$  are independent,

$$p(x, y) = p_X(x)p_Y(y)$$

Substituting,

$$p_Z(z) = \sum_x p_X(x)p_Y(z - x)$$

This is called a **convolution**.

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## Why Convolution Matters

Many important distributions arise from sums:

- Binomial
- Negative Binomial
- Poisson
- Gamma
- Normal

Convolution is the mathematical tool behind them.

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## Continuous Case

Suppose

$$Z = X + Y$$

where  $X, Y$  are continuous.

Then

$$f_Z(z) = \int_{-\infty}^{\infty} f(x, z-x) dx$$

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# Convolution for Independent Variables

If independent,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

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This is the continuous convolution formula.

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# Expectation of a Sum

One of the most powerful properties.

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## Linearity of Expectation

$$E[X + Y] = E[X] + E[Y]$$

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# Important

This is true even when:

- $X$  and  $Y$  are dependent

No independence is needed.

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# General Form

$$E \left[ \sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

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## Variance of a Sum

For independent random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

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More generally,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

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This shows why covariance is important.

Dependence affects variance.

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## Summary of Part 2

### Conditional Distribution

Discrete:

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Continuous:

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

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# Independence

$$P(X = x|Y = y) = P(X = x)$$

when  $X$  and  $Y$  are independent.

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## Conditional Expectation

Discrete:

$$E[X|Y = y] = \sum_x xP(X = x|Y = y)$$

Continuous:

$$E[X|Y = y] = \int x f_{X|Y}(x|y) dx$$

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## Law of Total Expectation

$$E[E(X|Y)] = E(X)$$

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## Sum of Random Variables

Discrete:

$$p_Z(z) = \sum_x p(x, z - x)$$

Independent case:

$$p_Z(z) = \sum_x p_X(x)p_Y(z - x)$$

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## Continuous Convolution

$$f_Z(z) = \int f_X(x)f_Y(z - x) dx$$

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# Key Takeaway

Part 2 introduces three foundational ideas:

1. **Conditional distributions** tell us how probabilities change when new information is revealed.
2. **Conditional expectation** gives the best average prediction after observing another variable.
3. **Convolution and sums** allow us to build new distributions from existing random variables, a technique that appears throughout probability, statistics, stochastic processes, and finance.